Which Lines Overload, and Which Neighbourhoods Lose Power

A linear-programming reliability study of O‘ahu’s 138 kV transmission grid, run on one recorded year and on 10,000 random draws

Abstract

We rebuild O‘ahu’s 138 kV transmission network from public records — 11 substations, 16 corridors carrying 25 circuits, and the island’s 24 dispatchable generating units (1,508 MW) — and run one optimisation model on it: a **linear program** that finds the cheapest way to meet every substation’s demand without pushing any line past its rating. Demand at each substation is the utility’s filed 2021 hourly island load, split between substations by census population and jobs. Each substation’s 8,760 hourly values are summarised by a **Gaussian distribution** — a mean and a standard deviation — and the eleven Gaussians are joined by their measured correlations. We solve the program on the 8,760 recorded hours and on 10,000 random hours drawn from those Gaussians, first with every line in service and then with every single circuit (N-1) and every pair of circuits (N-2) removed, the test the utility’s own planning rule sets.

With all lines in service, nothing goes wrong: no line is ever more than 73 % full and all demand is served in every recorded hour and every random draw. **Losing any one circuit still loses no demand**, but it pins the two single circuits that carry the west side’s power east (CEIP–Ewa Nui and Kalaeloa–Ewa Nui) at their limit in 27 % to 39 % of hours. **Losing two circuits cuts power in 7 of 128 cases**: East Honolulu loses everything if both Koolau–Pukele circuits fail; downtown Honolulu is short in 99 % of hours if two of its three supply circuits fail; Wahiawā and the North Shore are short in 37 % of hours if both of their lines fail; and the windward side with East Honolulu is short in 8 % of hours if two of the three lines into Koolau fail. Three of these — East Honolulu, downtown and the windward side — are the weak spots the utility itself described to the Hawai‘i Public Utilities Commission in 2007. The random draws agree on every one of them and add three rare cases (at most 0.33 % of draws) that the recorded year never produced, because the Gaussian produces demands, and mixes of demands, that the recorded year never had. Because the substation split is built from three island-wide sector curves, its covariance matrix is singular; giving every substation its own independent local variation makes it full rank and leaves these answers in place, except that Pearl Harbor becomes exposed in the recorded year too.

How to read this paper. Every number the model uses is either *measured* (taken from a public record of O‘ahu’s own grid) or a *stand-in* (taken from the closest documented source when O‘ahu’s own value is not public). Stand-ins are marked where they are used, and appendix A lists every input with its source. Sources are cited with short tags such as [HIFLD] or [PUC]; the tags are resolved in the reference list.

1 The optimisation problem, and the network it runs on

1.1 What is being decided

Every hour, the grid operator answers one question: *which power stations run, and how hard?* Many answers keep the lights on, and they cost different amounts. The job is to find the cheapest one that is physically legal.

Think of the network as reservoirs joined by pipes. Three rules apply.

Nothing is stored.
At each substation, power in equals power out, in the same instant.

You cannot steer the flow.
How much power goes down each line is not a choice. It is set by the difference in voltage phase angle between the two ends — the electrical equivalent of a pressure difference. Water takes every open pipe at once, in proportion to how easy each pipe is; so does power.

Every pipe has a width.
Each line has a rating in megawatts. Its flow, in either direction, must stay inside it.

1.2 The variables

Symbol	Units	What it is, and its physical picture
Given to the model — fixed numbers		
d_n	MW	the demand at substation n this hour: how much water the town at reservoir n drinks. *This is the only input that changes between experiments.*
$\bar{p}_g$	MW	the maximum output of generator g: the size of its pump
c_g	\$/MWh	the running cost of one megawatt-hour from generator g (fuel plus variable maintenance): what each pump charges per litre
b_ℓ	per unit	the susceptance of line ℓ: how *easy* the pipe is. A short line, or several circuits side by side, is a wide, short pipe with a large b_ℓ
$\bar{F}_\ell$	MW	the thermal rating of line ℓ: the most the pipe can carry before it overheats
V	10,000 \$/MWh	the penalty for each megawatt-hour of demand that is not served — a fine so large that the model will run every pump on the island before it lets a town go dry[MISO]
Chosen by the model — the decision variables		
$p_g \geq 0$	MW	how hard generator g is run
θ_n	radians	the voltage phase angle at substation n: the water level in reservoir n
$s_n \geq 0$	MW	demand that cannot be served at substation n — the town that goes without
Follows from the above		
F_ℓ	MW	the flow on line ℓ

With 11 substations and 24 generators the program has $24 + 11 + 11 = 46$ decision variables.

1.3 The linear program

$$\min_{p,\theta,s} \quad \underbrace{\sum_g c_g\, p_g}_{\text{running cost}} \quad + \quad \underbrace{V \sum_n s_n}_{\text{penalty for unserved demand}} \tag{1}$$

subject to, for every substation n and every line ℓ:

$$F_\ell = 100\, b_\ell \left(\theta_{f(\ell)} - \theta_{t(\ell)}\right) \qquad \text{flow follows the angle difference} \qquad (2)$$

$$\sum_{g \text{ at } n} p_g \;+\; s_n \;-\; \sum_{\ell \text{ out of } n} F_\ell \;+\; \sum_{\ell \text{ into } n} F_\ell = d_n \qquad \text{power in = power out} \qquad (3)$$

$$-\bar{F}_\ell \;\le\; F_\ell \;\le\; \bar{F}_\ell \qquad \text{the line's rating, both directions} \qquad (4)$$

$$0 \;\le\; p_g \;\le\; \bar{p}_g \qquad \text{the generator's range} \qquad (5)$$

$$0 \;\le\; s_n \;\le\; d_n \qquad \text{only existing demand can go unserved} \qquad (6)$$

$$\theta_{\text{Waiau}} = 0 \qquad \text{one angle fixes the zero point} \qquad (7)$$

The "100" in eq. (2) is the 100 MVA per-unit base, the usual convention[MAT]: it converts per-unit susceptance times radians into megawatts. Equation (2) is the standard *DC power flow*: it keeps the physics that decides where power goes (the angle rule) and drops the physics that decides voltages and losses, which is the usual trade for planning studies of this kind.

Every term is a variable times a constant, so this is a *linear program*. A linear program has no local minima to get trapped in: the solver (HiGHS, called from JuMP in Julia) either proves it has found the cheapest answer or proves there is none. Every one of the 19,930,640 programs solved for this paper returned a proven optimum.

A note on the word "overload". In this model a line can never exceed its rating: eq. (4) is a wall, and the optimiser does not walk through walls. What it does instead is press right up against one, and re-route power the expensive way around. So in this paper **a line "overloads" when its flow reaches its rating** — the point at which, without re-dispatch, it would be pushed past it. When re-dispatching is no longer enough, the model leaves demand unserved ($s_n > 0$), and that is the separate, more serious failure: **an area "loses power"**.

Losing a line. To test an outage we re-solve the same program with that line's circuit removed. A corridor of k identical circuits side by side has k times the susceptance and k times the rating of one circuit; losing one of them leaves $(k-1)/k$ of both. Losing the last one removes the corridor. The generators are re-dispatched after the outage, as an operator would.

1.4 The network

Figure 1 shows the network. It was assembled from four kinds of public record:

- **Which substations exist and where:** the federal HIFLD map of transmission lines[HIFLD]. Each 138 kV line in it names the two substations it joins; a substation sits where all of its lines meet.
- **How many circuits join each pair:** the utility's single-line diagrams in its 2021 Renewable Energy Zone study[REZ], which name existing circuits one by one ("Kahe–Halawa #1", "Waiau–Makalapa 2"); a worked example in its IGP interconnection appendix[RFP] ("Waiau-Ewa Nui 1 and Waiau-Ewa Nui 2 138kV circuits"); and the Public Utilities Commission's 2007 order on the East O'ahu Transmission Project[PUC] ("Two 138kv transmission lines currently feed the Pukele Substation from the Koolau Substation").
- **How long each line is:** measured along the HIFLD route itself.
- **Where each power plant connects:** its own switchyard (Kahe, Waiau), the nearest 138 kV substation (Schofield at Wahiawā), or — for H-Power, Kalaeloa and CIP CT-1 — the AES switching station, whose line the utility describes as carrying power "from existing

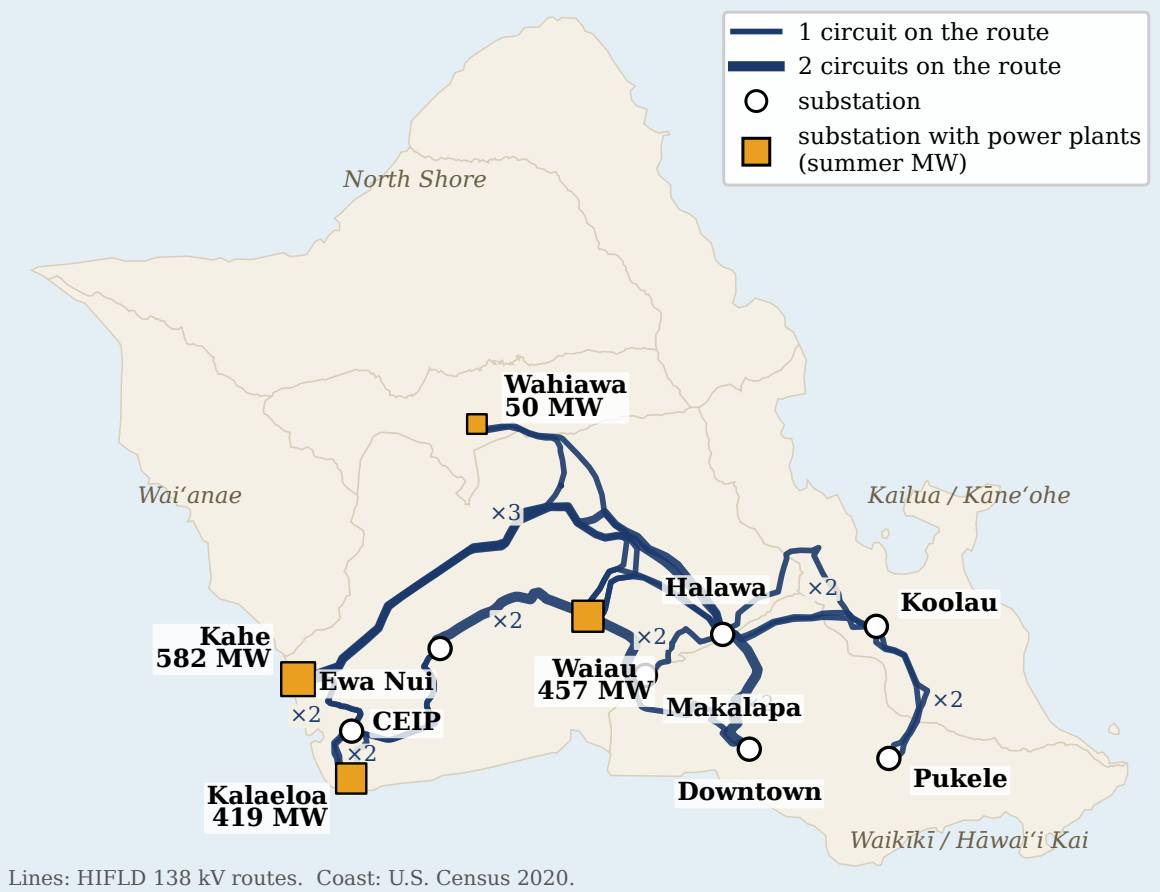


Figure 1: O‘ahu’s 138 kV transmission network as modelled. Blue lines are the real 138 kV routes from the federal HIFLD map[HIFLD]; a thicker line carries two circuits on one route, and “$\times n$” gives the number of circuits in a corridor. Squares are substations with power plants attached, sized by their summer capacity (EIA-860[EIA860]). The coastline is the Census 2020 shoreline[CB].

independent power producers AES, H-Power, and Kalaeloa, as well as from Hawaiian Electric’s CIP CT-1 generating unit”[AES].

Three simplifications are made. The downtown substations — Iwilei and School Street, which the Commission names as the two serving downtown, and Archer, Kewalo and Kamoku, which it also mentions[PUC] — are merged into one bus, *Downtown*: HIFLD draws only one of them (Kewalo), and no public source shows how they are wired to each other. The three circuits that feed downtown from outside — “three 138kv transmission lines providing power from the Halawa and Makalapa Substations”[PUC] — are kept. The AES switching station and the Kalaeloa plant, 0.46 mi apart on the HIFLD map, are merged into one bus, *Kalaeloa*. And the 46 kV sub-transmission network below the 138 kV system is not modelled (section 6).

Two numbers are stand-ins. The utility does not publish the reactance or the rating of its existing lines, so both come from the closest documented source.

- **Rating: 209.9 MW per circuit.** In its 2021 study the utility draws the new 138 kV circuits it would build into Halawa as single “795 AAC” conductors (Figs. 30, 33, 36), and when it re-conductors existing lines it specifies “double-bundled 795 AAC” (Table 5-3)[REZ] — so the existing lines it would upgrade carry less than two such conductors today. We use one 795 kcmil all-aluminium conductor per circuit. Its manufacturer rates

Table 1: The 16 corridors. Rating is per corridor (circuits × 209.9 MW). Circuit counts: [REZ], [RFP], [AES] and [PUC] as listed; otherwise the number of HIFLD routes.

Corridor	Circuits	Miles	Rating (MW)	Circuit count from
Kahe–CEIP	2	4.26	419.7	HIFLD; [AES] ("Kahe-CEIP 1&2")
CEIP–Kalaeloa	2	2.15	419.7	HIFLD (two routes); [AES]
Kalaeloa–Ewa Nui	1	8.29	209.9	HIFLD
CEIP–Ewa Nui	1	6.58	209.9	HIFLD
Ewa Nui–Waiau	2	7.21	419.7	[REZ] Fig. 34; [RFP]
Kahe–Waiau	1	18.81	209.9	HIFLD
Kahe–Halawa	3	20.69	629.6	[REZ] Figs. 30, 33 (with Kahe–Makalapa)
Kahe–Wahiawa	1	17.82	209.9	HIFLD; [REZ] Fig. 32 (Akau–Wahiawa)
Wahiawa–Waiau	1	12.23	209.9	[REZ] Fig. 32, Table 5-3
Waiau–Makalapa	2	4.52	419.7	[REZ] Fig. 34
Waiau–Koolau	2	13.28	419.7	[REZ] Table 10-3 (#1, #2); [PUC]
Halawa–Koolau	1	9.71	209.9	[REZ] Figs. 30, 31; [PUC]
Halawa–Makalapa	1	4.26	209.9	HIFLD
Halawa–Downtown	2	6.26	419.7	[REZ] Fig. 30 (Iwilei, School St)
Makalapa–Downtown	1	7.09	209.9	[PUC] (third downtown circuit)
Koolau–Pukele	2	6.32	419.7	[PUC]; [REZ] Fig. 31
Total	**25**			

it at 878 A[SW] (75 °C conductor, 25 °C air, 2 ft/s wind, in sun), and one three-phase circuit then carries $\sqrt{3} \times 138\,\text{kV} \times 878\,\text{A} = 209.9\,\text{MVA}$. Picture the rating as the width of the pipe: a thicker wire carries more current before it heats up and sags. For one corridor this is probably low: the utility's 2016 application to move part of the AES–CEIP 2 circuit specifies a 1,780 kcmil conductor for the new section[AES]. Because the true ratings are unknown, every result is repeated with 150 MW per circuit (three times the surge impedance loading of a typical 138 kV line, the usual limit for short lines[UT]) and with 318.6 MW (the 1,590 kcmil conductor the utility labels on its planned new Kahe–Wahiawa and Makalapa circuits[REZ], rated 1,333 A[SW]; the same study's tables print "1950 AAC" for some of these circuits, which is not a standard size); see section 4.6.

- **Reactance: 0.48 Ω/km**, the typical value for a 138 kV line with one conductor per phase[UT], the same quantity synthetic-grid builders compute from conductor geometry[GEG]. Because every line gets the same ohms per mile, this number *cancels out*: power divides between parallel paths in proportion to 1/length whatever the ohms per mile. Only the lengths, which are measured, matter.

1.5 The generators

The fleet is the island's 24 dispatchable units as reported to the U.S. Energy Information Administration for 2024[EIA860], 1,508.1 MW in total. Each unit's running cost is its measured heat rate[EIA923] times the utility's 2024 fuel-price forecast[IGP], plus a variable maintenance cost. That cost is a stand-in: the utility publishes values only for *new* combined-cycle, combustion-turbine and engine plants[IGP], which we use for its existing ones, and none at all for steam or waste-to-energy units, for which we use the U.S. EPA's values for existing units[EPA]. It is small next to fuel and changes no unit's place in the order in which plants are run. H-Power burns the island's municipal waste and is given no fuel cost, which makes it the first plant to run.

Plant	Substation	MW	$/MWh
H-Power (waste-to-energy)	Kalaeloa	86.0	6.94
Kalaeloa (combined cycle)	Kalaeloa	220.0	94.05
Kahe 1–6 (oil steam)	Kahe	582.1	122.28
Waiau 3–8 (oil steam)	Waiau	354.2	130.36
Schofield (engines)	Wahiawa	50.4	173.33
Waiau 9–10 (combustion turbines)	Waiau	102.4	224.84
CIP CT-1 (combustion turbine)	Kalaeloa	113.0	240.47
Total		**1508.1**	

Utility-scale solar, wind and batteries are not in this fleet; the demand the fleet must meet is the island load net of rooftop solar (section 2). The fleet therefore has to cover the whole net load by itself, which makes the model conservative about supply.

2 How the substation statistics are obtained

2.1 Demand at each substation, hour by hour

The model needs d_n for every substation in every hour. No public record gives it, so it is built in three steps from records that do exist.

1. **The island total.** The utility's filed hourly load for 2021 (Integrated Grid Planning inputs, Workbook 3[IGP]), 8,760 values. It is the load the grid must deliver: customer demand minus rooftop solar, customer batteries and energy efficiency, each an hourly layer in the same workbook. This total is kept exactly.
2. **Who is where.** Every 2020 census tract (people[CEN]) and every census block with jobs (LEHD LODES 2021[LODES]) is given to its nearest load-serving 138 kV substation — each substation serves its own "catchment", the rule the open European grid model PyPSA-Eur uses for the same purpose[PYPSA]. The Koʻolau mountains are respected: a neighbourhood is served only from a substation on its own side of the range (the thinly populated north-east shore, Koʻolauloa, may also be served around the north end, from Wahiawā). The utility's list of the neighbourhoods fed from Pukele ("Kahala, Kaimuki, Manoa, Makiki, and Waikiki"[PUC]) is followed and extended east to Hawaiʻi Kai: census tracts 1–35, which run from Hawaiʻi Kai west to Makiki, and any tract east of 157.830° W go to Pukele.
3. **What kind of load, hour by hour.** The island's sales are split into homes, businesses and industry by the utility's 2021 sales[EIA861]; each sector gets its own daily shape (U.S. DOE reference buildings for Honolulu[OEI]; flat for industry). Homes follow people and businesses follow jobs. Rooftop solar and efficiency are subtracted where their own sector shares[EIA861] say they are.

The same three steps as one equation. Picture three buckets of paint — homes, businesses, industry. Every hour each bucket holds a different amount: the island's load in that sector. Every substation pours from the same three buckets with its own *fixed recipe*. For substation b in hour t:

$$d_b(t) \;=\; k_{b,\mathrm{res}}\, N_{\mathrm{res}}(t) \;+\; k_{b,\mathrm{com}}\, N_{\mathrm{com}}(t) \;+\; k_{b,\mathrm{ind}}\, N_{\mathrm{ind}}(t), \tag{8}$$

or, for all ten substations with load at once,

$$\underbrace{D}_{10\times 8760} \;=\; \underbrace{K}_{10\times 3}\;\underbrace{N}_{3\times 8760}\,.$$

Symbol	Material object	Where it comes from
$N_s(t)$	how full **bucket** s is in hour t: the island's load of sector s, in MW	the filed island total, split by 2021 sales shares[EIA861] and hourly shapes[OEI], minus that sector's share of rooftop solar, efficiency and batteries
$k_{b,s}$	substation b's **recipe**: its share of the island's people (homes), commercial jobs (businesses) or industrial jobs (industry)	census 2020[CEN] and LODES 2021[LODES]; the same in every hour
$d_b(t)$	what substation b pours in hour t, in MW	the product

Over 2021 the three buckets average 147 MW (homes; below zero at sunny middays, when rooftop solar exceeds home use), 215 MW (businesses) and 382 MW (industry). The recipes are:

	share of island (%)				share of island (%)		
Substation	people	com. jobs	ind. jobs	**Substation**	people	com. jobs	ind. jobs
Downtown	13.0	46.2	41.6	Ewa Nui	12.7	3.6	5.9
Pukele	21.6	19.6	6.2	Wahiawa	12.4	4.6	5.3
Makalapa	6.0	9.3	14.1	Koolau	11.9	5.3	4.7
CEIP	4.1	3.1	12.2	Kahe	5.6	1.7	1.2
Waiau	9.6	6.0	8.0	Halawa	3.2	0.7	0.7

The result is an $11 \times 8{,}760$ table of megawatts (Kalaeloa, a switching station, has no load). Its one public check is three substation shares the utility gave the Commission in 2007[PUC]: Pukele "supplied electricity to approximately seventeen percent of the Oahu load" at the 2002 day peak (our table: 14.1 % at the 2021 day peak); Koolau and Pukele together serve "approximately thirty percent" (no date given; ours: 20.3 %); and Iwilei, School Street and the Honolulu Power Plant served "about twenty-five percent" at the 2002 day peak (ours: 38.4 %, for a Downtown bus that also holds Archer, Kewalo and Kamoku, Kaka'ako and Ala Moana). The filed shares are nineteen years older and partly cover different substations, so they are not a target; but because they differ, every result is repeated with the table re-weighted to match them (section 4.6).

2.2 The Gaussian model of each substation

Each substation now has 8,760 numbers. We summarise them with a **Gaussian (normal) distribution**, the bell curve:

$$d_n \;\sim\; \mathcal{N}\left(\mu_n,\, \sigma_n^2\right), \qquad f(x) = \frac{1}{\sigma_n\sqrt{2\pi}}\,\exp\!\left(-\frac{(x-\mu_n)^2}{2\sigma_n^2}\right),$$

$$\mu_n = \frac{1}{8760}\sum_{h=1}^{8760} d_{n,h}, \qquad \sigma_n^2 = \frac{1}{8759}\sum_{h=1}^{8760}\left(d_{n,h}-\mu_n\right)^2.$$

Picture a sandpile. Pour one grain for each hour of the year onto a ruler marked in megawatts, at the demand that hour had. μ_n is where the pile balances; σ_n is how wide it spreads — about two thirds of the grains of a true bell-shaped pile lie within one σ_n of the middle.

Figure 2 is honest about the fit. The bell has the right middle and the right width, but the recorded hours are two-humped: 61 % of Downtown's hours and 65 % of Pukele's lie within one σ of the mean, against 68 % for a true bell. The Gaussian is a summary, not a copy.

Table 2: The Gaussian of each substation. μ and σ from the 8,760 hours of 2021. σ/μ strips out size and leaves only how much a substation swings. Share: share of the island at the 2021 day peak (the 50 heaviest hours between 10:00 and 16:00).

Substation	Serves	μ (MW)	σ (MW)	σ/μ	Share
Downtown	downtown, Kalihi, Kaka'ako, Ala Moana	277.2	35.3	0.13	38.4%
Pukele	Waikīkī, Mānoa, Kaimukī, Kāhala, Hawai'i Kai	97.6	26.6	0.27	14.1%
Makalapa	Pearl Harbor, Hickam, Salt Lake	82.5	9.1	0.11	10.8%
CEIP	Campbell Industrial Park, Kapolei	59.5	5.3	0.09	7.3%
Waiau	Pearl City, Waipahu, Mililani	57.4	10.8	0.19	7.5%
Ewa Nui	Ēwa, Ēwa Beach	48.9	13.6	0.28	6.2%
Wahiawa	Wahiawā, Schofield, the North Shore	48.2	13.4	0.28	6.2%
Koolau	Kāne'ohe, Kailua, the windward coast	47.0	13.1	0.28	6.2%
Kahe	the Wai'anae coast	16.6	6.0	0.36	2.2%
Halawa	'Aiea, Hālawa	8.8	3.4	0.39	1.1%
Kalaeloa	— (switching station)	0	0	—	—
Island		**743.9**	**124.9**	**0.17**	

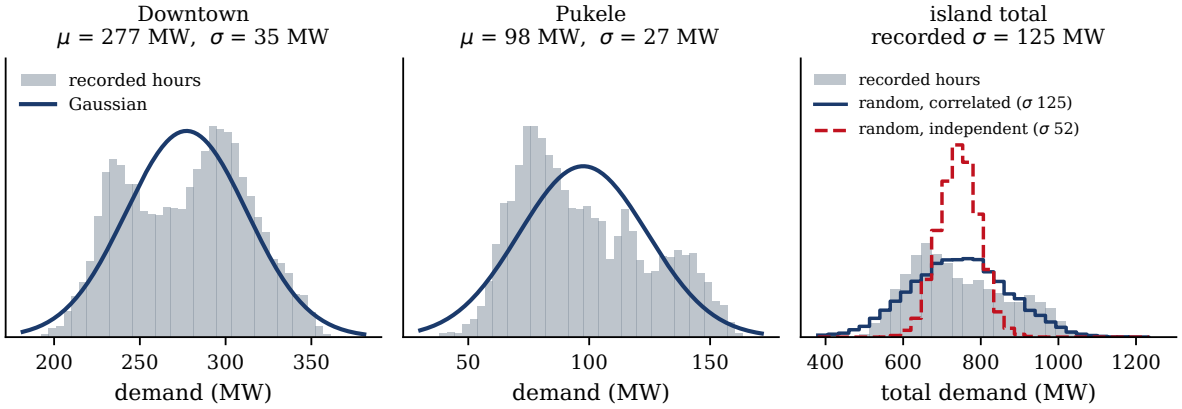


Figure 2: The bell curves behind the random draws. *Left and centre:* grey bars are the 8,760 recorded hours at two substations; the blue curve is the Gaussian drawn from only the μ and σ above each panel. The curve gets the centre and the spread right but not the shape: real demand has two humps, a night hump and a day hump. *Right:* the island total. Correlated draws (blue) have the same spread as the recorded year; independent draws (red dashes) have well under half of it.

2.3 The substations move together

Drawing each substation from its own bell curve *separately* gives eleven correct substations and a wrong island. The island total is a sum, and a sum of independent terms cancels most of their swings. Real substations do not cancel: when O'ahu is busy it is busy everywhere at once. Over the recorded year every pair of substations is positively correlated (from 0.47 to 1.00, average 0.86).

Demand input	Mean	Std. dev.	Highest	Hours above 1,000 MW
the recorded year (8,760 hours)	743.9 MW	124.9 MW	1,054.3 MW	1.83 %
eleven bells drawn *separately* (10,000)	743.6 MW	52.0 MW	960.2 MW	0.00 %
eleven bells drawn *together* (10,000)	743.0 MW	125.1 MW	1,216.0 MW	1.89 %

So the random draws use the **multivariate Gaussian**: the same eleven means, plus the measured 11×11 covariance matrix Σ, which records how every pair of substations moves together:

$$\mathbf{d} \;=\; \max(\mathbf{0},\, \boldsymbol{\mu} + L\,\mathbf{z})\,, \qquad \mathbf{z} \sim \mathcal{N}(\mathbf{0},\, I_{11}), \qquad LL^{\top} = \Sigma.$$

Picture eleven corks on one pond. $\mathbf{z}$ is a handful of random waves; L is the shape of the pond, which decides how each wave lifts every cork at once. Each cork still bobs exactly as much as its own σ_n says, but now they rise and fall together as the real substations do. The draw is floored at zero because demand cannot be negative.

Why Σ is singular, and what we do about it. Because $D = KN$, every substation's demand is a mix of the same three bucket curves, so the ten substations with load have only three independent directions of change: $\Sigma = K\,\Sigma_N K^{\top}$ has rank three, with eigenvalues 2,295, 452 and 17 MW^2 and seven exact zeros. Two things follow. Any substation is an exact mix of three others: Pukele $=$ 0.338 Downtown $+$ 1.875 Wahiawā $-$ 1.456 CEIP in every hour, to within 0.0002 MW. And two substations with nearly the same recipe are nearly proportional: Koolau stays between 0.95 and 1.00 times Wahiawā all year (correlation 0.999).

Real substations do not behave like this. Each also moves on its own — local weather and clouds over rooftop solar, one hospital or base changing its load, switching — so metered data would give a full-rank, if highly correlated, Σ. The singularity is a property of how our demand table is built, not of O'ahu. It has a technical consequence: L comes from an eigendecomposition of Σ, because the usual Cholesky factorisation fails on a singular matrix. It also has a substantive one: the draws can never produce one substation rising on its own. So every outage test is repeated with a **local wiggle** added to each substation (section 4.7):

$$\mathbf{d} \;=\; \max(\mathbf{0},\, \boldsymbol{\mu} + L\,\mathbf{z} + f\,\boldsymbol{\sigma} \odot \boldsymbol{\eta})\,, \qquad \boldsymbol{\eta} \sim \mathcal{N}(\mathbf{0}, I), \qquad \mathrm{Cov}(\mathbf{d}) \approx \Sigma + f^2\,\mathrm{diag}(\sigma_n^2). \tag{9}$$

The waves $\mathbf{z}$ still lift every cork together; $\boldsymbol{\eta}$ gives each cork its own small independent bob, f times its usual swing σ_n. The added diagonal makes the covariance full rank. The same wiggle is added to the recorded hours.

One consequence matters for the results. The Gaussian's tails are longer than the recorded year's: the random draws reach 1,216 MW, 15 % above the recorded peak of 1,054 MW, and 0.56 % of draws exceed that peak. The random input is therefore a stress test of the year, not a replay of it.

3 The experiments

The same linear program is solved again and again. Between solves only two things change: the demand vector $\mathbf{d}$, and which lines are in service.

Demand input	
recorded	the 8,760 hourly demand vectors of 2021, one solve per hour
random	10,000 vectors drawn from the multivariate Gaussian of section 2, one solve per draw
Network state	
N-0	every circuit in service (1 case)
N-1	one circuit out, each in turn (16 distinct cases: the circuits of a corridor are identical, so losing either of two is the same case)
N-2	two circuits out, every pair (128 distinct cases). This is the utility's own planning test: "no transmission component shall exceed its emergency rating with one generating unit on overhaul, one transmission line out for maintenance and loss of a second transmission line"[PUC]
growth	every line in service, every demand multiplied by $g = 1.0, 1.1, \ldots, 2.0$
Local variation	
$f = 0.10, 0.25, 0.50$	N-0, N-1, N-2 and growth repeated with each substation's own wiggle added (eq. (9)), in every recorded hour and random draw

Each outage case is solved on all 8,760 recorded hours *and* all 10,000 random draws, so every percentage below is a share of *all* hours or *all* draws, not of a sample. Two remarks on the outage test. It uses normal ratings, because emergency ratings are not published; since a real line may run above its normal rating for a few hours after a second failure, our N-2 results are the conservative (pessimistic) side of the utility's test. And the generators are allowed to re-dispatch after the outage, as they would be.

4 Results

4.1 All lines in service: nothing goes wrong

With every circuit in service, **all demand is served in all 8,760 recorded hours and all 10,000 random draws, and no line ever reaches its rating** (fig. 3). The fullest line is CEIP–Ewa Nui, at 72.6 % in its heaviest recorded hour and 73.0 % in the heaviest random draw; the next are Kalaeloa–Ewa Nui (68 %) and Halawa–Downtown (62 % recorded, 67 % random). This is consistent with the utility's own 2021 power-flow study, which found "no existing system equipment overloading issue" in the conditions it studied[REZ].

The two fullest lines have the same job: they carry the west side's power *east* — all of the output of the island's two cheapest plants, H-Power and the Kalaeloa combined cycle (306 MW between them, run in every hour), plus part of Kahe's. Everything generated west of Ewa Nui except what goes north on the Kahe lines has to cross one of these two single circuits.

The random draws load the lines a little more than the recorded year (for Waiau–Makalapa, 61 % against 52 %) because their tails reach higher demand, but they never reach a rating either.

4.2 One circuit lost (N-1): no demand lost, two lines pinned

Removing any one of the 25 circuits, in any recorded hour or any random draw, **never leaves demand unserved**. What changes is that some lines are pushed to their rating, and the optimiser has to re-dispatch — run dearer plant elsewhere — to keep them there:

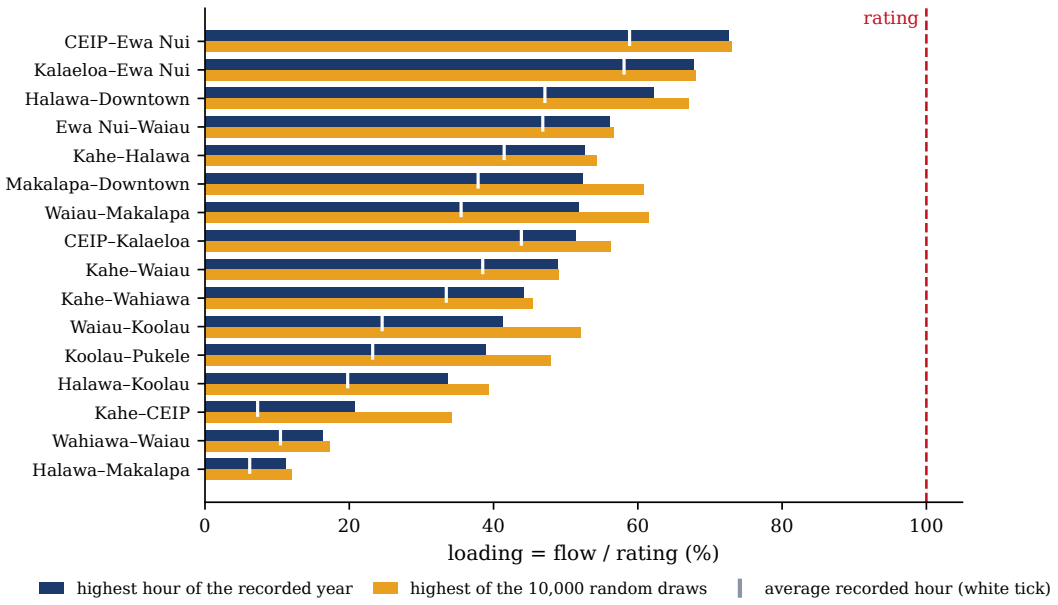


Figure 3: How full each line gets with every line in service. Loading is flow divided by rating. Blue: the heaviest hour of the recorded year; gold: the heaviest of the 10,000 random draws; white tick: the average recorded hour. No line reaches its rating (red dashes) in any hour or any draw.

Circuit lost	Line pushed to its rating	Recorded hours	Random draws
Kalaeloa–Ewa Nui	CEIP–Ewa Nui	3,401 (38.8 %)	4,235 (42.4 %)
CEIP–Ewa Nui	Kalaeloa–Ewa Nui	2,359 (26.9 %)	2,523 (25.2 %)
one Halawa–Downtown circuit	the other one	1 (0.01 %)	32 (0.32 %)
one CEIP–Kalaeloa circuit	the other one	0	17 (0.17 %)
one Waiau–Makalapa circuit	the other one	0	4 (0.04 %)

The first two rows are the export pair again. Lose one of the two single circuits east of Kalaeloa and the other is full for a quarter to two fifths of the year. The optimiser does not touch the cheap Kalaeloa plants (they stay at 306 MW); it runs Kahe less — Kahe's power also reaches the full line, through CEIP — and Waiau, which is east of the bottleneck, more. With Kalaeloa–Ewa Nui out, Kahe gives up 61 MW on average in the pinned hours and Waiau adds the same, at an extra running cost of $491 per pinned hour. The system still meets the utility's single-outage test, but the margin that makes it pass is *generation that can be moved*, not spare wire.

4.3 Two circuits lost (N-2): four areas can lose power

Of the 128 ways to lose two circuits, **7 leave demand unserved in the recorded year, and 10 in the random draws** (table 3 and fig. 4). In the recorded year they fall into four areas, and each has one simple reason: it hangs off two or three circuits, and two failures leave it with too little wire, or none. The random draws add a fifth, rarely: Pearl Harbor (Makalapa).

East Honolulu (Pukele) — the largest single exposure.
Pukele serves Waikīkī, Mānoa, Kaimukī, Kāhala and Hawai'i Kai and is fed at 138 kV by exactly two circuits, both from Koolau[PUC]. Lose both and there is no 138 kV path at all: the whole Pukele load, up to 163 MW, is unserved in every hour. The utility told the Commission the same: if both lines are out, "approximately ninety-three percent of the customers serviced from the Pukele Substation will experience an outage"[PUC]. The 2007 order approved a 46 kV project that "will allow load to be shifted among the three substations using

Table 3: The double outages that leave demand unserved. Hours: share of the 8,760 recorded hours with any unserved demand; draws: share of the 10,000 random draws. MWh: total unserved over the recorded year. Worst: the largest unserved demand in any single recorded hour. "One X circuit" means one of the parallel circuits of corridor X.

Circuits lost	Area short	Hours	Draws	MWh / yr	Worst
both Koolau–Pukele circuits	East Honolulu	100%	99.99%	854,579	163 MW
both Halawa–Downtown circuits	Downtown	98.9%	97.4%	590,276	160 MW
one Halawa–Downtown circuit + Makalapa–Downtown	Downtown	98.9%	97.4%	590,709	160 MW
Kahe–Wahiawa + Wahiawa–Waiau	Wahiawā, North Shore	36.8%	43.1%	41,750	29 MW
both Waiau–Koolau circuits	Windward + East Honolulu	7.9%	4.9%	6,893	31 MW
one Waiau–Koolau circuit + Halawa–Koolau	Windward + East Honolulu	7.9%	4.9%	6,927	31 MW
one Waiau–Makalapa + one Halawa–Downtown circuit	Downtown	0.01%	0.53%	5	5 MW
Only in the random draws					
one Kahe–Halawa + one Waiau–Makalapa circuit	Pearl Harbor (Makalapa)	0	0.33%	0	—
one Waiau–Makalapa + one Waiau–Koolau circuit	Pearl Harbor (Makalapa)	0	0.08%	0	—
Halawa–Koolau + one Halawa–Downtown circuit	Downtown	0	0.01%	0	—

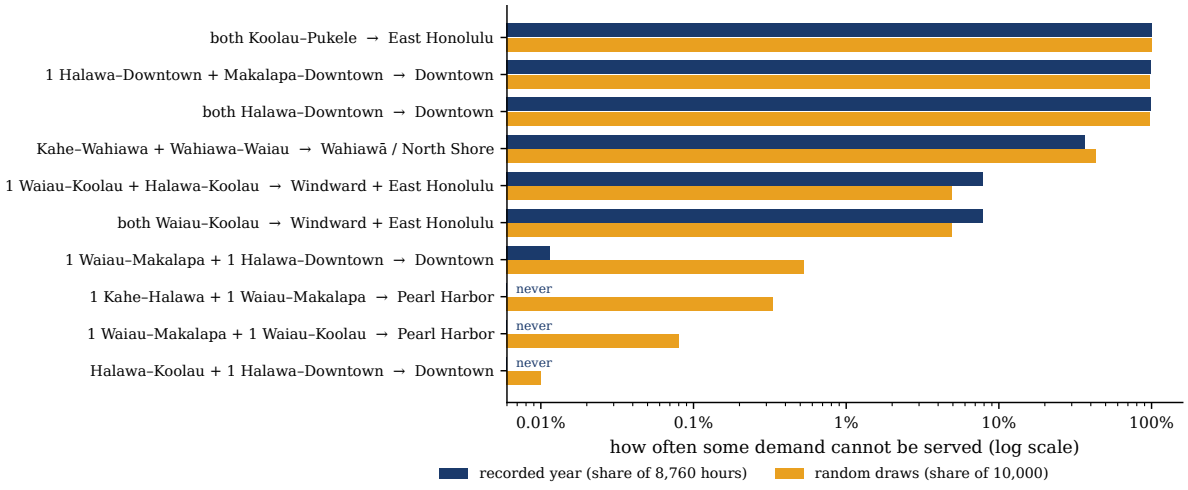


Figure 4: Two circuits lost: how often each area cannot be fully served. Every double outage that leaves demand unserved in either input; "1 X" means one of the parallel circuits of corridor X, "both X" means both. Log scale: each gridline is ten times the one before. Blue, the recorded year; gold, the random draws.

46kv lines"[PUC] — Pukele, Archer and Kamoku. That 46 kV backstop is outside our model, so this row is what the 138 kV network alone would do.

Downtown Honolulu — the utility's own "Downtown Overload Situation".
Downtown is fed by three circuits (two from Halawa, one from Makalapa). Lose any two and the last one carries at most 209.9 MW, while downtown's demand runs from 192 MW to 370 MW and is above 209.9 MW in 8,661 of the 8,760 hours — exactly the hours in which demand goes unserved (99 %). The utility described the same event to the Commission: if one downtown line is out for maintenance "and a second Downtown 138kv transmission line becomes unavailable, then the current flowing through the remaining Downtown 138kV transmission line will exceed the emergency current carrying capacity rating"[PUC]. It expected this from 2007 without the Honolulu Power Plant downtown[PUC]; that plant is no longer in the fleet[EIA860].

Wahiawā and the North Shore.
Wahiawā has two 138 kV lines, one to Kahe and one to Waiau[HIFLD] (the utility's diagram names them Akau–Wahiawa and Waiau–Wahiawa[REZ]). Lose both and it becomes an island with only the 50.4 MW Schofield plant, which covers its demand in the lighter 63 % of hours and falls short, by up to 29 MW, in the other 37 %.

The windward side and East Honolulu together.
Koolau (and Pukele behind it) is fed by three circuits[PUC]. Lose two and the third carries the whole windward-plus-East-Honolulu load alone; in the heaviest 8 % of hours it cannot. The utility projected exactly this: with one Koolau line out for maintenance and a second unavailable, "the current flowing through the third 138kV Koolau transmission line will exceed its emergency current carrying capacity rating during daytime peak load conditions"[PUC].

Real against random. The two inputs agree on all seven recorded failures, and within about six percentage points on how often they happen. Where they differ, the random draws are reaching demands the recorded year never had: an island total above the recorded peak, one substation above its own recorded maximum, or a mix of substation demands the year never produced. That is why the random input finds three extra rare cases (Pearl Harbor twice, downtown once), each in at most 0.33 % of draws and never in the recorded year, and why it makes the downtown "one Waiau–Makalapa + one Halawa–Downtown" case about 46 times more frequent (0.53 % of draws against one recorded hour, 0.011 %). For the windward rows the order flips (4.9 % of draws against 7.9 % of hours). Those failures happen whenever Koolau and Pukele together need more than the 210 MW one circuit can carry, and the real year's upper hump puts more hours just above that line (7.87 %) than the bell curve does (4.87 %): the Gaussian has the longer tail, the real year the fatter shoulder.

Against the utility's own study. The utility's 2021 power-flow study checked normal, single-outage and double-outage conditions (the last at emergency ratings) for the dispatches it studied and found "no existing system equipment overloading issue"[REZ]. Our model finds demand lost after some double outages. The difference is what the model leaves out: emergency ratings, the real line ratings, and the 46 kV network that can move load between substations. So the rows of table 3 show *where the 138 kV network is thinnest*; they are not a prediction that customers there would lose power for that long.

4.4 Which lines overload the most

Table 4 ranks every line by how often it is pushed to its rating across all outage cases. Read a cell "*c* / *h*" as: in *c* of the outage cases this line reached its rating, for *h* hours (or draws) in total

across those cases.

Table 4: Every line, ranked by how often it reaches its rating. N-0: highest loading with every line in service. N-1 and N-2: number of outage cases in which the line reaches its rating / total hours (recorded) or draws (random) at the rating, summed over those cases. "—": never.

	N-0 highest		N-1 (16 cases)		N-2 (128 cases)	
Line	rec.	rand.	rec.	rand.	rec.	rand.
CEIP–Ewa Nui	73%	73%	1 / 3,401	1 / 4,235	14 / 50,548	14 / 62,724
Kalaeloa–Ewa Nui	68%	68%	1 / 2,359	1 / 2,523	14 / 48,068	14 / 56,284
Halawa–Downtown	62%	67%	1 / 1	1 / 32	13 / 9,061	15 / 11,113
CEIP–Kalaeloa	51%	56%	—	1 / 17	2 / 8,767	14 / 10,371
Makalapa–Downtown	52%	61%	—	—	1 / 8,663	1 / 9,743
Ewa Nui–Waiau	56%	57%	—	—	3 / 4,002	3 / 4,955
Waiau–Koolau	41%	52%	—	—	1 / 702	1 / 503
Halawa–Koolau	34%	39%	—	—	1 / 698	1 / 501
Waiau–Makalapa	52%	61%	—	1 / 4	3 / 83	8 / 295
Kahe–Halawa	53%	54%	—	—	—	—
Kahe–Waiau	49%	49%	—	—	—	—
Kahe–Wahiawa	44%	45%	—	—	—	—
Koolau–Pukele	39%	48%	—	—	—	—
Kahe–CEIP	21%	34%	—	—	—	—
Wahiawa–Waiau	16%	17%	—	—	—	—
Halawa–Makalapa	11%	12%	—	—	—	—

Three groups stand out.

- **The export pair — CEIP–Ewa Nui and Kalaeloa–Ewa Nui — overloads the most**, by a wide margin: they are the two fullest lines in normal operation, the only lines pushed to their rating by a single outage in a large share of hours, and each is pushed to it in 14 of the 128 double outages. The cause is where the plants are — the cheapest power on the island enters at Kalaeloa and must leave by these two single circuits — and the heavier the island's load, the more often they fill: after a single outage the remaining one is full in every hour when the island needs more than 833 MW, and in none below 756 MW.
- **The three downtown feeders** — two circuits from Halawa and one from Makalapa — matter most for customers. Halawa–Downtown reaches its rating in 13 double outages. When a downtown feeder fills, the only relief is the other downtown feeders, because downtown has no plant of its own; that is why these are the overloads that end in lost demand. Under growth, Makalapa–Downtown and Waiau–Makalapa are the first lines to fill.
- **Seven lines never reach their rating in any case**: Kahe–Halawa, Kahe–Waiau, Kahe–Wahiawa, Koolau–Pukele, Kahe–CEIP, Wahiawa–Waiau and Halawa–Makalapa. Koolau–Pukele is on this list even though losing it is the worst outage on the island: it never overloads, it simply disappears.

4.5 Demand growth: the generators run out before the wires

To separate a shortage of *wire* from a shortage of *generation*, every growth case is solved twice: once as usual, and once with all line limits removed (a "copper plate"). Demand lost in the copper plate is a generation shortage; demand lost only when the limits are on is caused by the wires.

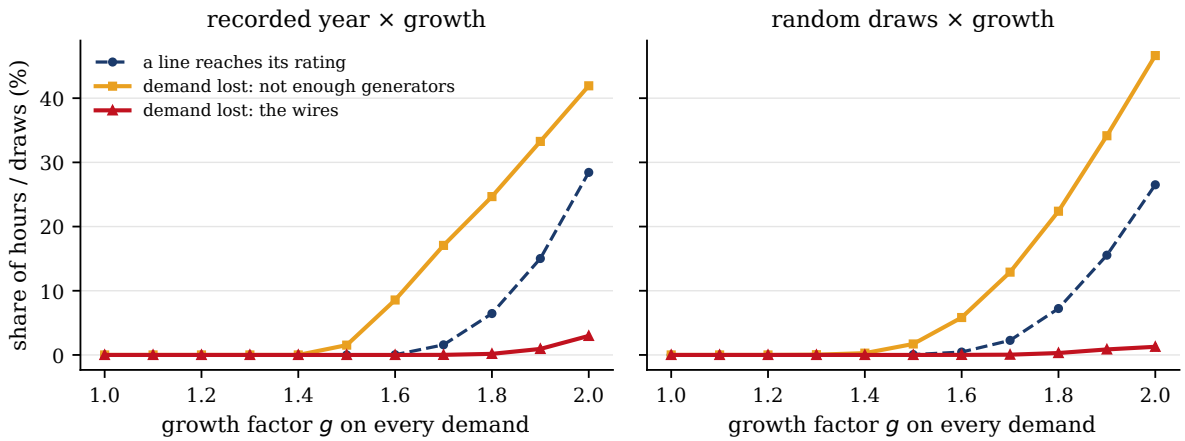


Figure 5: Every demand multiplied by g, every line in service. Blue dashes: share of hours (left) or draws (right) in which some line reaches its rating. Gold: demand lost because the island's generators together are too small (it is lost even with the line limits removed). Red: demand lost because of the wires alone (served when the line limits are removed).

First happens at growth factor …	Recorded year	Random draws
generators too small in some hour (fleet 1,508 MW)	$g = 1.5$	$g = 1.3$
a line reaches its rating	$g = 1.6$	$g = 1.5$
…the lines: Waiau–Makalapa, Makalapa–Downtown		
demand lost because of the wires	$g = 1.7$	$g = 1.6$
…the areas: Pearl Harbor (Makalapa) and Downtown		

Figure 5 shows the result: **with all lines in service, the recorded year's demand can grow by 40 % before anything runs short (the random draws, with their longer tails, by 20 %), and what runs short first is generation, not wire.** Even at twice today's demand the wires alone cost power in only 3.0 % of recorded hours and 1.3 % of random draws, against 42 % and 47 % for the generators. The random draws reach each threshold one or two steps earlier because their tails are longer. The Kalaeloa export pair never reaches its rating under growth (CEIP–Ewa Nui peaks at 95 % of it in the recorded year and 97 % in the random draws): its flow is set by the plants at Kalaeloa, which already deliver their 306 MW in every recorded hour and add only the last-resort CIP turbine as demand grows.

4.6 How much the answers depend on the stand-in numbers

Two inputs are stand-ins with real uncertainty: the line ratings, and the split of island demand between substations. Every outage case was re-run with each alternative.

Table 5 separates what is robust from what is not.

- **Robust:** no variant loses demand with every line in service. East Honolulu, downtown and Wahiawā can lose power in every variant, and the windward side in all but the thickest-line one; Pearl Harbor joins only with the thinnest lines or in the random draws. East Honolulu's exposure does not depend on any stand-in at all: with no 138 kV path left, the rating is irrelevant. Wahiawā's depends only on the Schofield plant's size and Wahiawā's demand.
- **Not robust:** how *often* downtown and the windward side are short. Thicker lines (318.6 MW) cut downtown's exposure from 99 % to 13 % of hours and remove the windward one. The 2002 split moves load from downtown to the windward side and East Honolulu, and the two swap places: downtown falls to 10 %, the windward side

Table 5: Sensitivity. Recorded-year results (random draws in brackets) for three per-circuit ratings and for the substation split re-weighted to the 2002 filed shares (Pukele 17%, Koolau 13%, Downtown 25%, rest in proportion)[PUC].

	150 MW per circuit	**209.9 MW (base) per circuit**	**318.6 MW per circuit**	**209.9 MW, 2002 shares**
N-0: highest loading	100% (100%)	73% (73%)	48% (48%)	74% (75%)
N-0: demand lost	never	never	never	never
N-1: cases that lose demand (of 16)	4 (7)	0 (0)	0 (0)	0 (1)
N-2: cases that lose demand (of 128)	60 (91)	7 (10)	4 (4)	6 (21)
Share of hours short after the four key double outages				
both Koolau–Pukele → East Honolulu	100%	100%	100%	100%
both Halawa–Downtown → Downtown	100%	98.9%	12.6%	10.0%
Kahe–Wahiawa + Wahiawa–Waiau → Wahiawā	36.8%	36.8%	36.8%	46.1%
both Waiau–Koolau → Windward + E. Honolulu	39.2%	7.9%	0%	45.9%

rises to 46 %. (That variant gives the whole merged Downtown bus the 25% filed for Iwilei and School Street alone, so it probably understates downtown.)

- **The 150 MW rating is very unlikely for the real system.** With it, losing a single downtown circuit would leave downtown short in 30 % of hours — a failure of the utility's own single-outage rule that its 2021 study, which found "no existing system equipment overloading issue"[REZ], would not have missed.

4.7 Local variation: does a full-rank Σ change the answers?

The demand table has no substation-specific variation, so its covariance matrix is singular (section 2). Table 6 repeats every outage test with the local wiggle of eq. (9) at three sizes: $f = 0.10$, 0.25 and 0.50 of each substation's own swing σ_n. Nobody publishes the right size — it would take metered substation data — so these bracket it. Picture the cloud of demand vectors: without the wiggle it is a flat three-dimensional sheet in a ten-dimensional room; the wiggle gives it thickness in every direction.

- **The covariance becomes full rank, and the island barely notices.** The wiggles are independent, so across the island they mostly cancel: the island's spread rises by at most 2.7 MW. What changes is how tightly substations move together — at $f = 0.5$ Wahiawā and Koolau are no longer near-copies (0.80 instead of 0.999).
- **The four at-risk areas and their order do not change.** No variant loses demand with every line in service, and no recorded hour loses demand after a single outage. The four areas' shortfall frequencies move by at most 3.2 percentagepoints.
- **Pearl Harbor is the area that local variation exposes.** Makalapa, which hangs mostly off the two Waiau–Makalapa circuits, went short only in random draws before; with a wiggle of a quarter of its swing or more it goes short in recorded hours too (15 hours at $f = 0.5$).
- **The extra random-draw cases are rare.** Of the 37 double outages that lose demand in some draw at $f = 0.5$ (against 10 without the wiggle), 21 do so in only one or two of the 10,000 draws — mostly East Honolulu with one of its two circuits out, in a draw where Pukele alone rises above what one circuit carries.

Table 6: Adding local variation. Recorded-year results, random draws in brackets. f: size of each substation's independent wiggle as a share of its own standard deviation. Thinnest direction: square root of the smallest eigenvalue of the recorded-hour covariance matrix (zero means singular).

	$f = 0$ (base)	$f = 0.10$	$f = 0.25$	$f = 0.50$
What the wiggle does to the demand				
thinnest direction of the demand cloud	0 MW	0.37 MW	0.94 MW	1.86 MW
average correlation between substations	0.86	0.85	0.81	0.69
Wahiawā–Koolau correlation	0.999	0.989	0.941	0.802
spread of the island total, σ	124.9 MW	125.0 MW	125.6 MW	127.6 MW
What it does to the answers				
N-0: demand lost	never	never	never	never
N-1: cases that lose demand (of 16)	0 (0)	0 (0)	0 (0)	0 (3)
N-2: cases that lose demand (of 128)	7 (10)	7 (11)	8 (14)	9 (37)
both Koolau–Pukele → East Honolulu	100%	100%	100%	100%
both Halawa–Downtown → Downtown	98.9%	98.7%	98.0%	95.7%
Kahe–Wahiawa + Wahiawa–Waiau → Wahiawā	36.8%	37.0%	37.9%	40.0%
both Waiau–Koolau → Windward + E. Honolulu	7.9%	8.0%	8.4%	8.6%
1 Kahe–Halawa + 1 Waiau–Makalapa → Pearl Harbor	0 h (33)	0 h (36)	1 h (44)	15 h (65)
growth: generators first short at	$g = 1.5$ (1.3)			$g = 1.4$ (1.3)
growth: wires first cost demand at	$g = 1.7$ (1.6)			$g = 1.7$ (1.5)

5 Answers

Question	Answer (base case; recorded year, random draws)
How often can the network not supply the demand?	With every line in service: **never** — 0 of 8,760 hours, 0 of 10,000 draws. After any one circuit fails: never. After two circuits fail: in 7 of the 128 possible pairs (10 in the random draws); in those, from one hour a year to every hour.
Which areas are most at risk?	**1. East Honolulu** (Pukele: Waikīkī, Mānoa, Kaimukī, Kāhala, Hawai'i Kai) — two circuits, one source; losing both cuts everything on the 138 kV network. **2. Downtown Honolulu** — short in 99 % of hours after losing two of its three feeders. **3. Wahiawā and the North Shore** — islanded on the 50 MW Schofield plant, short in 37 % of hours (43 % of draws). **4. The windward side with East Honolulu** — short in 8 % of hours (5 % of draws) after losing two of the three lines into Koolau. Pearl Harbor (Makalapa) is at risk in the random draws, under growth, and — once each substation is given its own local variation — in recorded hours too.
Which lines overload the most?	**CEIP–Ewa Nui and Kalaeloa–Ewa Nui**, the two single circuits that carry the west side's power east (all of Kalaeloa's, part of Kahe's): the fullest lines in normal operation (73 % and 68 %), and pinned at their rating in 39 % and 27 % of hours when the other one is out. Then the **downtown feeders**, whose overloads are the ones that end in lost demand. Under growth, **Waiau–Makalapa and Makalapa–Downtown** fill first (at 1.6× demand).
Does random demand give the same answers as the recorded year?	**Yes on every major finding** — the same areas, the same lines, the same order, and frequencies within about six percentage points. The random draws add three rare failures (at most 0.33 % of draws) that the recorded year never shows, and reach each growth threshold one or two steps earlier, because they include demands the year never had. They are more pessimistic about rare events, not about every row: the windward failure is *less* frequent in the draws.

The broad lesson is about *shape*, not size. O'ahu's 138 kV grid has ample room in normal operation and survives any single failure. Its exposure is concentrated in four places where a large load hangs off two or three circuits with no plant of its own. Three of the four are the weaknesses the utility itself described to its regulator in 2007[PUC]. The match is partly built in — the circuit counts for those areas come partly from the same order — but the model adds what the order does not give: how often, and by how much, each area would fall short on the 138 kV network.

6 Limitations

- **Stand-in ratings and reactances.** The utility does not publish them. The reactance cancels out (section 1.4); the rating does not, and table 5 shows how far it moves the downtown and windward results.
- **No 46 kV network.** Below 138 kV, O'ahu has a 46 kV network that can move some load between neighbouring substations — the 2007 order approved exactly such ties for Pukele[PUC]. An area that "loses power" here might, in reality, lose part of it, or lose it only until switching is done.
- **Normal ratings for double outages.** The utility's rule allows emergency ratings after a second failure. Those are not published, so the N-2 results are pessimistic.
- **No metered substation data.** Hawaiian Electric does not publish hourly load by substation, so the substation split is built from census data by eq. (8), and its shares differ from the only filed ones, from 2002 (section 2). Metered data would replace three stand-ins at once: the split, the correlations between substations, and each substation's own local variation. The local-noise runs (section 4.7) only bracket how much the last one matters.
- **A three-factor demand.** Because every substation is a fixed mix of three island-wide sector curves, the substations move together more tightly than real ones do (correlations 0.47 to 1.00) and Σ has rank three.
- **The industrial class.** Industry is 42 % of the utility's 2021 sales[EIA861] (51 % of net load after rooftop solar, which falls mostly on homes). It is given a flat daily shape and placed by industrial jobs; which large customers the class contains, and where they are, is not public.
- **The Gaussian is a summary.** It matches each substation's mean, spread and correlations, not its two-humped shape, and its tails reach demands the recorded year never had.
- **DC power flow, hour by hour.** Voltages, reactive power and losses are ignored, and each hour is solved on its own, with no start-up, minimum-output or ramping limits. Generator outages (the "one generating unit on overhaul" in the utility's rule) are not tested.
- **Fleet and year.** The fleet is the 2024 fossil and waste fleet; the demand is 2021's. Utility-scale solar, wind and batteries are not modelled, so the fleet must meet the whole net load alone.

7 Reproducing this

Every number in this paper is computed by the scripts in this folder, in this order, and `check_numbers.py` confirms that each number typed in the text matches them.

`build_network.py`	substations, corridors, lengths, ratings (from HIFLD)
`build_generators.py`	the 24 units, their bus and running cost
`build_demand.py`	hourly demand at each substation
`build_demand_puc.py`	the same, re-weighted to the 2002 filed shares
`run_all.jl`	N-0 on the recorded hours and the random draws; Gaussian statistics
`contingency.jl`	N-0, N-1 and N-2 (set `RATING=low|high`, `DEMAND=puc` or `NOISE=0.25` for the sensitivities)
`stress.jl`	the growth ladder (also with `NOISE`)
`noise_stats.jl`	what the local wiggle does to the demand inputs
`redispatch.jl`	what the optimiser does when an export line is lost
`summarize.py`	the numbers and tables in the text
`check_numbers.py`	recomputes the quoted numbers and checks the text
`makefigs.py`	the five figures

In total the runs solve 19,930,640 linear programs; every one returned a proven optimum.

A Every input, and where it comes from

Measured means taken from a public record of O‘ahu’s own system. *Stand-in* means O‘ahu’s own value is not public, so the closest documented value is used, and the text says what it stands in for.

Input	Value used	Source	Kind
Substations and their locations	11 buses	HIFLD transmission lines, line end points[HIFLD]	measured
Which substations are joined	16 corridors	HIFLD[HIFLD]; single-line diagrams[REZ]; PUC order[PUC]	measured
Circuits per corridor	1–3, 25 in all	[REZ] Figs. 30–36, Table 10-3; [RFP]; [AES]; [PUC]; HIFLD	measured
Line lengths	2.15 mi to 20.69 mi	measured along HIFLD routes[HIFLD]	measured
Line reactance	0.48 Ω/km	typical 138 kV value[UT]; cancels from flows	stand-in
Per-circuit rating	209.9 MW	795 AAC, as drawn for new O‘ahu circuits[REZ]; 878 A[SW]	stand-in
Rating, low	150 MW	3× SIL of a 138 kV line[UT]	stand-in
Rating, high	318.6 MW	1590 AAC, [REZ] Figs. 32, 34, 35; 1,333 A[SW]	stand-in
Power base	100 MVA	MATPOWER convention[MAT]	convention
Generating units, capacity	24 units, 1,508.1 MW	EIA-860 2024, summer capacity[EIA860]	measured
Heat rates	per plant	EIA-923 2024[EIA923]	measured
Fuel prices	2024 forecast	utility IGP Workbook 3[IGP]	utility forecast
Variable O&M: CC, CT, engines	$2, $7, $31/MWh	utility IGP, Tables A-9, A-10 (2024)[IGP]	stand-in (new units)
Variable O&M: steam, waste	$0.88, $6.94/MWh	EPA Platform v6, Table 4-8 (2019$)[EPA]	stand-in
H-Power fuel cost	$0/MWh	burns municipal waste	assumption
Plant-to-substation map	4 buses	own switchyards; [AES]; nearest substation	measured

Input	Value used	Source	Kind
Penalty for unserved demand	$10,000/MWh	MISO value of lost load, FERC-accepted 2025[MISO]	stand-in
Island hourly load, 2021	8,760 values	utility IGP Workbook 3[IGP]	measured (filed)
Population	2020 tracts	U.S. Census 2020[CEN]	measured
Jobs	2021 blocks	LEHD LODES8[LODES]	measured
Sector sales, PV and efficiency shares	2021	EIA-861[EIA861]	measured
Hourly shapes	homes, 15 building types	DOE reference buildings, Honolulu[OEI]; CBECS 2018[CBECS]	stand-in
Catchment rule	nearest substation	PyPSA-Eur[PYPSA]; Pukele list from [PUC]	method
Filed substation shares	17%, 30%, 25%	PUC order[PUC] (17% and 25% at the 2002 day peak)	filed (2002)
Local variation per substation	$f = 0.10, 0.25, 0.50$ of σ_n	none published; metered substation load is not public	sensitivity
Coastline	2020 shoreline	Census cartographic boundaries[CB]	measured

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